

SIMATS ENGINEERING

CO4 ASSESSMENT TOOL 3

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Q13. Dice Throw Problem vs Coin Change — DP Comparison

Problem Statement

Given n dice, each having m faces numbered from 1 to m , find the number of ways to obtain a target sum X . Compare this problem with the Coin Change problem using Dynamic Programming.

1. Dice Throw Problem

Input:

- n = number of dice
- m = number of faces on each die
- X = target sum

Goal: Find the number of ways to obtain sum X .

DP State

Let:

$dp[i][j]$ = number of ways to obtain sum j using exactly i dice.

Base Case

$$dp[0][0] = 1$$

There is exactly one way to obtain sum 0 using zero dice.

$$dp[0][j] = 0 \text{ for } j > 0.$$

Recurrence

For every possible face value k :

$$dp[i][j] = \sum dp[i-1][j-k]$$

where:

$1 \leq k \leq m$ and $j-k \geq 0$.

Algorithm

DiceThrow(n, m, X)

1. Create $dp[n+1][X+1]$
 2. Initialize all values to 0
 3. $dp[0][0] = 1$
 4. for $i = 1$ to n :
 5. for $j = 1$ to X :
 6. for $k = 1$ to m :
 7. if $j \geq k$:
 8. $dp[i][j] = dp[i][j] + dp[i-1][j-k]$
 9. return $dp[n][X]$
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2. Coin Change Problem

Given n coin denominations and a target amount A , find the number of ways to make the amount.

Let:

$dp[i][j]$ = number of ways to make amount j using the first i coins.

Recurrence

If $coin[i] \leq j$:

$dp[i][j] = dp[i-1][j] + dp[i][j-coin[i]]$

Otherwise:

$dp[i][j] = dp[i-1][j]$

The two choices are:

- **Exclude** the current coin.

- **Include** the current coin.
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3. Similarities

1. Both problems use **Dynamic Programming**.
 2. Both contain **overlapping subproblems**.
 3. Previously calculated results are stored and reused.
 4. Both use a table to represent smaller subproblems.
 5. The final answer is obtained from a larger DP state.
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4. Differences

Feature	Dice Throw	Coin Change
Selection	Fixed number of dice	Variable number of coins
Values	1 to m	Given denominations
Repetition	Exactly n dice	Usually unlimited coins
State	Dice and sum	Coins and amount
Main decision	Select a face value	Include/exclude a coin
Recurrence	Sum over possible faces	Include + exclude
Constraint	Exactly n dice	Amount must be formed

5. Subproblem Overlap

In the Dice Throw problem, the same state such as $dp[2][5]$ can be required while calculating several larger states. Instead of recalculating it, DP stores its value.

Similarly, in Coin Change, the same amount can occur repeatedly for different combinations of coins.

Therefore, **memoization or tabulation reduces repeated computation**.

6. Complexity Analysis

Dice Throw

There are $n \times X$ states, and for each state we consider m face values.

Time Complexity:

$O(n \times X \times m)$

Space Complexity:

$O(n \times X)$

Coin Change

There are $n \times A$ states and each state takes constant time.

Time Complexity:

$O(n \times A)$

Space Complexity:

$O(n \times A)$

Space can be optimized to:

$O(A)$

using a one-dimensional DP array.

7. Scalability

For larger values of n , m , and X , the Dice Throw problem requires more computation because every state may examine all m faces. Coin Change is generally more scalable when the number of coin denominations and target amount are manageable.

Space optimization using a **1D DP array** can significantly reduce memory usage in both problems.

8. DP Modeling Insight

The important steps in modeling a DP problem are:

1. **Identify the state** — what information is required to describe a subproblem?
2. **Define the recurrence** — how is the current state obtained from smaller states?
3. **Identify base cases** — determine the smallest solvable problems.
4. **Find overlapping subproblems** — avoid repeated calculations.
5. **Analyze complexity** — determine time and space requirements.
6. **Optimize space** when only previous states are required.

Conclusion

The Dice Throw and Coin Change problems demonstrate how Dynamic Programming converts recursive problems with overlapping subproblems into efficient algorithms.

Although both use similar DP principles, their **states, constraints, and recurrence relations differ**. Dice Throw uses a fixed number of dice and considers every possible face, while Coin Change works with given denominations and usually allows repeated coin selection.
